

Workshop Management System

Final Report

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Project Overview

The Workshop Management System is a comprehensive platform designed to streamline the management and coordination of educational workshops. It serves three main user roles - Administrators, Instructors, and Participants - each with distinct functionalities tailored to their needs.

Core Purpose: The system facilitates the organization and delivery of workshops, handles session scheduling, manages participant registrations, and provides assessment capabilities through tests and certificates.

Key Stakeholders and Their Functions:

1. Administrators

- Central management of workshop content and information
- Instructor management and assignment
- Oversight of all workshop activities and participant engagement
- System-wide monitoring and control

2. Instructors

- Session scheduling and management
- Test creation and assessment
- Participant progress tracking
- Schedule management with conflict prevention

3. Participants

- Workshop discovery and registration
- Session attendance tracking
- Test participation and completion
- Certificate acquisition
- Personal schedule management

Primary System Features:

1. Workshop Management

- Creation and modification of workshop details
- Session scheduling with conflict detection
- Venue management and allocation

2. User Management

- Role-based access control
- User profile management
- Activity tracking

3. Session Management

- Scheduling system with conflict prevention
- Attendance tracking
- Resource allocation

4. Assessment System

- Test creation and administration
- Automated grading
- Certificate generation and distribution

5. Booking System

- Session reservation handling
- Conflict detection for both instructors and participants
- Capacity management

Expected Benefits:

1. Organizational Efficiency

- Streamlined workshop administration
- Reduced scheduling conflicts
- Automated process management

2. Enhanced User Experience

- Clear role-based interfaces
- Self-service capabilities for participants
- Automated certificate generation

3. Quality Control

- Structured assessment system
- Progress tracking
- Performance monitoring

This system aims to create a seamless experience for all stakeholders while maintaining organizational control and educational quality standards.

Database Description

The Workshop Management System database is designed to manage educational workshops, session scheduling, and participant management. The system uses MongoDB as the database with Prisma as the ORM, providing a comprehensive solution for workshop management and participant engagement.

Key Entities and Relationships

Admin

- **Represents:** System administrators
- **Attributes:**
 - id (PK): Unique identifier (ObjectId)
 - email: Unique email address
 - password: Hashed password
 - username: Unique username
 - firstName: First name of admin
 - lastName: Last name of admin
 - street: Street address
 - city: City of residence
 - state: State of residence
 - zip: Postal code
 - dob: Date of birth
 - phoneNo: Contact number
 - role: UserRole (ADMIN)
- **Relationships:**
 - Can have multiple password reset requests
 - Can have multiple username reset requests

Instructor

- **Represents:** Workshop instructors
- **Attributes:**
 - id (PK): Unique identifier (ObjectId)
 - email: Unique email address
 - password: Hashed password
 - username: Unique username
 - firstName: First name
 - lastName: Last name
 - street: Street address
 - city: City of residence
 - state: State of residence
 - zip: Postal code
 - dob: Date of birth
 - phoneNo: Contact number
 - ssn: Social Security Number
 - role: UserRole (INSTRUCTOR)
 - hasResetPassword: Password reset status
- **Relationships:**

- Can have multiple sessions
- Can have multiple password reset requests
- Can have multiple username reset requests

Participant

- **Represents:** Workshop attendees
- **Attributes:**
 - id (PK): Unique identifier (ObjectId)
 - email: Unique email address
 - password: Hashed password
 - username: Unique username
 - firstName: First name
 - lastName: Last name
 - street: Street address
 - city: City of residence
 - state: State of residence
 - zip: Postal code
 - dob: Date of birth
 - phoneNo: Contact number
 - role: UserRole (PARTICIPANT)
- **Relationships:**
 - Can have multiple registrations
 - Can have multiple test registrations
 - Can have multiple payments
 - Can have multiple password reset requests
 - Can have multiple username reset requests

Workshop

- **Represents:** Educational workshops
- **Attributes:**
 - id (PK): Unique identifier (ObjectId)
 - title: Workshop title
 - description: Workshop description
 - venue: Location of the workshop
 - startDate: Workshop start date
 - endDate: Workshop end date
 - capacity: Maximum participants
 - price: Workshop fee (optional)
 - workshopType: WorkshopType (PAID/FREE)
- **Relationships:**
 - Can have multiple sessions

Session

- **Represents:** Individual workshop sessions
- **Attributes:**
 - id (PK): Unique identifier (ObjectId)
 - title: Session title
 - description: Session description
 - date: Session date

- startTime: Session start time
- endTime: Session end time
- workshopId: Reference to workshop
- instructorId: Reference to instructor
- **Relationships:**
 - Associated with one workshop
 - Associated with one instructor
 - Can have multiple tests
 - Can have multiple registrations

Test

- **Represents:** Session assessments
- **Attributes:**
 - id (PK): Unique identifier (ObjectId)
 - title: Test title
 - description: Test description
 - date: Test date
 - startTime: Test start time
 - endTime: Test end time
 - venue: Test location
 - capacity: Maximum participants
 - questions: Array of test questions
 - status: TestStatus (PENDING/COMPLETED)
 - sessionId: Reference to session
- **Relationships:**
 - Associated with one session
 - Can have multiple test registrations

Registration

- **Represents:** Session enrollments
- **Attributes:**
 - id (PK): Unique identifier (ObjectId)
 - participantId: Reference to participant
 - sessionId: Reference to session
- **Relationships:**
 - Associated with one participant
 - Associated with one session
 - Can have one payment

Payment

- **Represents:** Transaction records
- **Attributes:**
 - id (PK): Unique identifier (ObjectId)
 - amount: Payment amount
 - method: PaymentMethod (CREDIT_CARD/DEBIT_CARD/CASH)
 - participantId: Reference to participant
 - reservationId: Reference to registration
- **Relationships:**
 - Associated with one participant

- Associated with one registration

Entity Name	Description
Admin	System administrators who manage the overall workshop system, including creating workshops and managing instructors. They have full control over the system's operations.
Instructor	Educational professionals who conduct workshop sessions. They can create and manage their sessions, create tests, and view participant registrations.
Participant	Users who attend workshops and take tests. They can register for sessions, take tests, and receive certificates upon completion.
Workshop	Educational events that contain multiple sessions. They have specific venues, capacities, and can be either paid or free.
Session	Individual classes or meetings within a workshop. Each session has a specific instructor, time slot, and can include tests.
Test	Assessments created by instructors for specific sessions. They include questions and have capacity limits for participants.
Registration	Records of participants enrolling in specific sessions. Links participants to their chosen sessions and associated payments.
Payment	Financial transactions for paid workshops. Tracks payment methods and amounts for session registrations.
PasswordReset	Manages password reset requests for all user types (Admin, Instructor, Participant). Includes tokens with expiration dates.
UsernameReset	Handles username reset requests for all user types. Includes tokens with expiration dates.
InstructorRegistrationRequest	Tracks applications from potential instructors to join the system. Includes personal and professional information.
TestRegistration	Records of participants registering for specific tests. Links participants to their test attempts.

Functionality

Administrator Functions

- **Workshop Management**
 - Create new workshops with details (title, description, venue, dates, capacity)
 - Edit existing workshop information
 - Set workshop pricing (free or paid)
 - Monitor workshop capacities and enrollments
- **Instructor Management**
 - Add new instructors to the system
 - Edit instructor information
 - Review and approve instructor registration requests
 - Assign instructors to workshops
 - Monitor instructor activities
- **System Oversight**
 - View all workshop sessions
 - Monitor participant registrations
 - Track system usage and activities
 - Access to all workshop statistics

Instructor Functions

- **Session Management**
 - Create new sessions within assigned workshops
 - Schedule sessions with conflict detection
 - View and manage personal session schedule
 - Update session details and content
- **Test Administration**
 - Create tests for sessions
 - Set test parameters (date, time, venue, capacity)
 - Add test questions
 - Monitor test registrations
 - Track test completion status
- **Participant Tracking**
 - View session registrations
 - Monitor participant attendance
 - Access participant test results
 - View session-specific statistics

Participant Functions

- **Workshop Access**
 - View available workshops
 - Browse session schedules
 - Access workshop details and information
 - Filter workshops by type (paid/free)
- **Registration Management**
 - Register for workshop sessions

- View personal schedule
 - Check for scheduling conflicts
 - Cancel registrations if needed
 - Make payments for paid workshops
- **Test Participation**
 - Register for available tests
 - Take tests during scheduled times
 - View test results
 - Access completion certificates
 - Track personal progress

General System Functions

- **User Authentication**
 - Secure login system
 - Password reset functionality
 - Username reset capability
 - Role-based access control
 - Reset password and reset username functionality
- **Profile Management**
 - Update personal information
 - Modify contact details
 - Change password
 - Update username
- **Payment Processing**
 - Handle multiple payment methods
 - Process workshop fees
 - Track payment history
 - Generate payment receipts
- **Scheduling System**
 - Automatic conflict detection
 - Schedule validation
 - Time slot management
 - Venue availability checking

User Roles

1. Administrator (Admin)

- **Primary Role:** Overall system management and administration
- **Access Level:** Full system access
- **Responsibilities:**
 - Workshop creation and management
 - Instructor account approval and management
 - System-wide monitoring and control
 - User management across all roles
 - Access to all system features and data
- **Special Privileges:**
 - Can modify any workshop or session
 - Can manage instructor accounts
 - Can view all system analytics
 - Has override capabilities for most system functions

2. Instructor

- **Primary Role:** Workshop session delivery and management
- **Access Level:** Limited to assigned workshops and sessions
- **Responsibilities:**
 - Creating and managing sessions
 - Test creation and administration
 - Monitoring participant progress
 - Managing session schedules
- **Special Privileges:**
 - Can create tests for their sessions
 - Can view registrations for their sessions
 - Can manage their session schedules
 - Access to participant data for their sessions

3. Participant

- **Primary Role:** Workshop attendance and participation
- **Access Level:** Basic user access
- **Responsibilities:**
 - Managing personal registrations
 - Completing workshop sessions
 - Taking tests
 - Maintaining personal schedule
- **Special Privileges:**
 - Can register for workshops and sessions
 - Can take tests
 - Can view personal progress
 - Access to personal certificates

4. System (Built-in Role)

- **Primary Role:** Automated system operations
- **Access Level:** System-level operations
- **Responsibilities:**

- Automated scheduling
- Conflict detection
- Notification management
- System maintenance
- **Special Functions:**
 - Handles automated emails
 - Manages session conflicts
 - Processes automated updates
 - Maintains system integrity

Key Role Interactions:

- Admins oversee both Instructors and Participants
- Instructors interact directly with Participants through sessions and tests
- Participants interact with the system primarily through registration and attendance
- System role supports all other roles through automated functions

Customer Audience

Customer Audience for the Workshop Management System:

Primary Target Audiences:

1. Educational Institutions

- Universities and colleges
- Technical training schools
- Professional development centers
- Continuing education programs
- Corporate training departments

2. Professional Training Providers

- Corporate training organizations
- Professional certification bodies
- Skills development centers
- Industry-specific training providers
- Vocational training institutes

3. Workshop Participants

- Working professionals seeking skill enhancement
- Students pursuing additional certifications
- Career changers looking for new skills
- Industry professionals needing updated certifications
- Individuals interested in personal development

4. Independent Instructors

- Subject matter experts
- Industry professionals offering training
- Freelance educators
- Professional trainers
- Specialized skill instructors

Target Market Characteristics:

1. Size and Scale

- Small to medium training centers (10-100 concurrent workshops)
- Large educational institutions (100+ concurrent workshops)
- Individual instructors managing multiple workshops
- Organizations with multiple training locations

2. Technical Requirements

- Basic computer literacy
- Internet connectivity
- Mobile device accessibility
- Basic digital documentation capabilities

3. Industry Sectors

- Information Technology
- Business Management
- Healthcare Education
- Professional Services

- Technical Training
- Arts and Creative Fields
- Personal Development

4. Geographical Scope

- Local training centers
- Regional educational institutions
- National training providers
- International educational organizations

Key Benefits for Each Audience:

1. For Organizations

- Streamlined workshop management
- Reduced administrative overhead
- Improved tracking and reporting
- Enhanced participant engagement
- Better resource utilization

2. For Instructors

- Efficient session management
- Simplified test administration
- Better participant tracking
- Reduced scheduling conflicts
- Enhanced teaching tools

3. For Participants

- Easy workshop discovery
- Simple registration process
- Clear progress tracking
- Convenient test taking
- Automatic certification

Market Positioning:

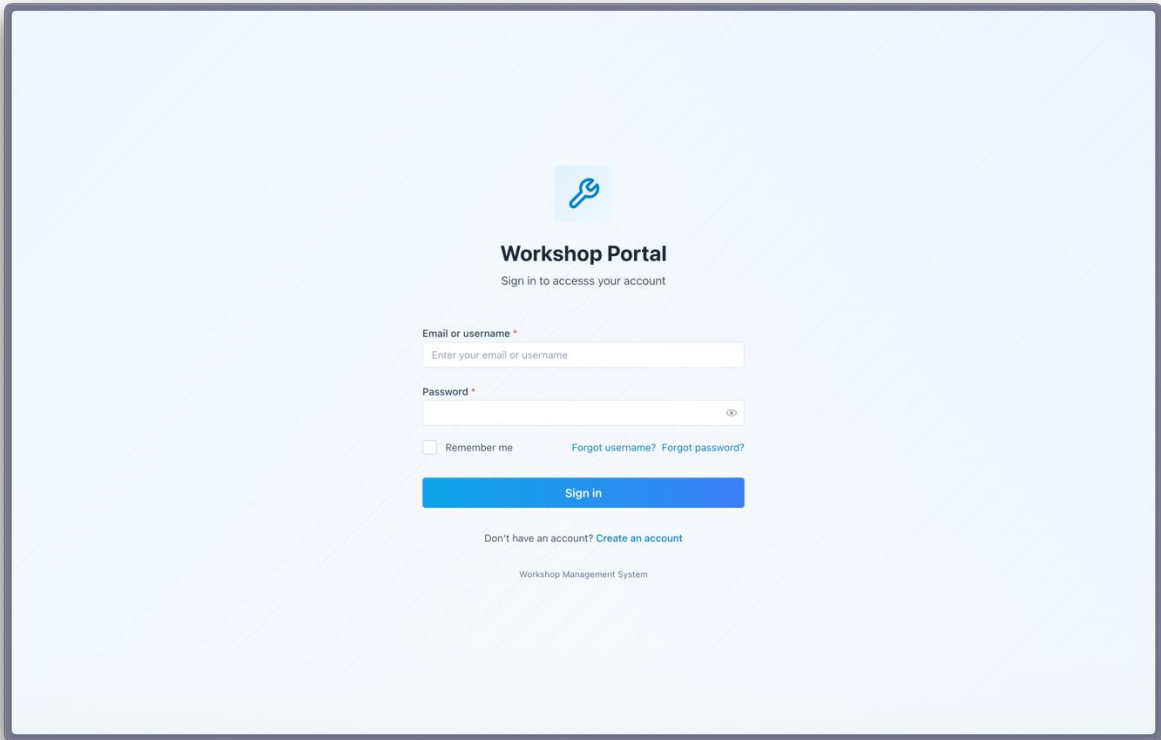
- Mid to high-end market segment
- Focus on quality and reliability
- Emphasis on user experience
- Professional and enterprise-grade solution
- Scalable for different organization sizes

Summary

The Workshop Management System is a comprehensive digital platform that revolutionizes the way educational workshops, training sessions, and certification programs are managed and delivered. Built with MongoDB and Prisma ORM, the system provides a robust infrastructure for handling complex workshop management tasks while ensuring data integrity and system reliability. The platform serves a diverse range of users, from educational institutions and corporate training departments to independent instructors and individual participants, offering tailored functionality for each user role.

At its core, the system operates on a three-tier user hierarchy consisting of Administrators, Instructors, and Participants, each with specifically designed access levels and capabilities. Administrators have full system control, managing workshops, instructors, and overseeing all operations. Instructors can create and manage sessions, develop tests, and track participant progress. Participants can browse workshops, register for sessions, take tests, and receive certificates upon completion. The platform includes sophisticated features such as intelligent scheduling with conflict detection, integrated payment processing for paid workshops, automated notification systems, and comprehensive reporting capabilities. Through its user-friendly interface and automated processes, the system significantly reduces administrative overhead while enhancing the overall learning experience. The platform's architecture ensures scalability and security, making it suitable for organizations of varying sizes and complexity levels. Whether it's managing a single workshop or coordinating multiple training programs across different venues, the system provides the necessary tools and functionality to streamline operations and improve educational delivery.

Screenshots



The screenshot shows the 'Workshop Portal' sign-in page. At the top center is a blue square icon with a white wrench. Below it, the title 'Workshop Portal' is displayed in bold, followed by the subtitle 'Sign in to access your account'. The form contains two input fields: 'Email or username' with a placeholder 'Enter your email or username' and 'Password' with a toggle icon. Below the password field are links for 'Remember me', 'Forgot username?', and 'Forgot password?'. A blue 'Sign in' button is centered below the form. At the bottom, there is a link 'Don't have an account? Create an account' and the text 'Workshop Management System'.

Workshop Portal
Sign in to access your account

Email or username *
Enter your email or username

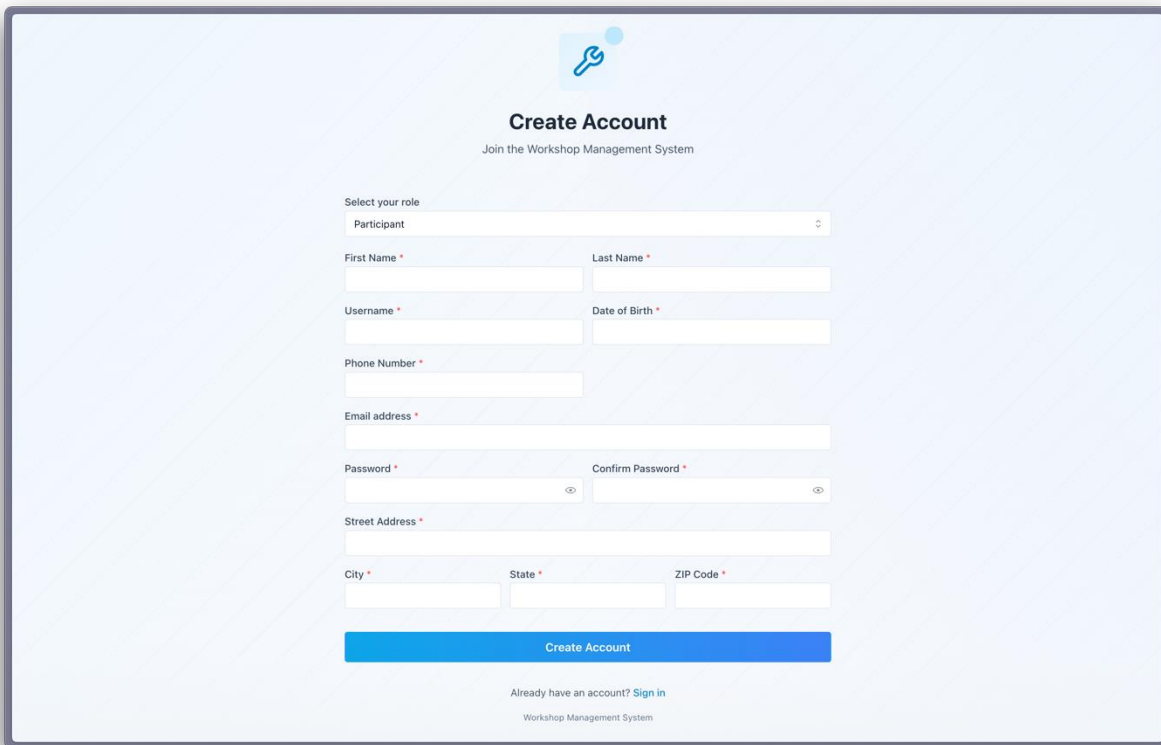
Password *
👁

☐ Remember me [Forgot username?](#) [Forgot password?](#)

Sign in

[Don't have an account? Create an account](#)

Workshop Management System



The screenshot shows the 'Create Account' page. At the top center is a blue square icon with a white wrench. Below it, the title 'Create Account' is displayed in bold, followed by the subtitle 'Join the Workshop Management System'. The form starts with a 'Select your role' dropdown menu set to 'Participant'. Below this are several input fields: 'First Name', 'Last Name', 'Username', 'Date of Birth', 'Phone Number', 'Email address', 'Password', 'Confirm Password', 'Street Address', 'City', 'State', and 'ZIP Code'. Each field has an asterisk indicating it is required. The 'Password' and 'Confirm Password' fields have toggle icons. A blue 'Create Account' button is centered below the form. At the bottom, there is a link 'Already have an account? Sign in' and the text 'Workshop Management System'.

Create Account
Join the Workshop Management System

Select your role
Participant

First Name * Last Name *

Username * Date of Birth *

Phone Number *

Email address *

Password * Confirm Password *
_____ _____

Street Address *

City * State * ZIP Code *
_____ _____ _____

Create Account

[Already have an account? Sign in](#)

Workshop Management System



Recover Username

We'll send you instructions to recover your username

Email *

Send Recovery Code

Remember your username? [Back to login](#)

Workshop Management System



Reset Password

We'll send you instructions to reset your password

Email *

Send Reset Code

Remember your password? [Back to login](#)

Workshop Management System

W Workshop

Instructors

Participants

Workshops

Requests

Reservations

JS John Smith
admin@app.com

Instructors

Manage your workshop instructors and their information

Add New Instructor

Name	Email	Phone	DOB	SSN	Gender
Praneetha Babburi	praneethababburi@gmail.com	5551234567	May 9, 1992	123456789	Edit
Sarah Davis	instructor2@app.com	5559876543	August 24, 1989	987654321	Edit

W Workshop

Instructors

Participants

Workshops

Requests

Reservations

JS John Smith
admin@app.com

Back to Instructors

Add New Instructor

First Name *

Last Name *

Email *

Password *

< >

Confirm Password *

< >

Phone Number *

SSN *

Date of Birth *

Street Address *

City *

State *

ZIP Code *

W Workshop

Instructors

Participants

Workshops

Requests

Reservations

JS John Smith
admin@app.com

< Back to Instructors

Edit Instructor

First Name *

Praneetha

Last Name *

Babburi

Email *

praneethababburi@gmail.com

Phone Number *

5551234567

SSN *

123456789

Date of Birth *

May 9, 1992

Street Address *

321 Trainer Blvd

City *

Miami

State *

FL

ZIP Code *

33101

Cancel

Update Instructor

W Workshop

Instructors

Participants

Workshops

Requests

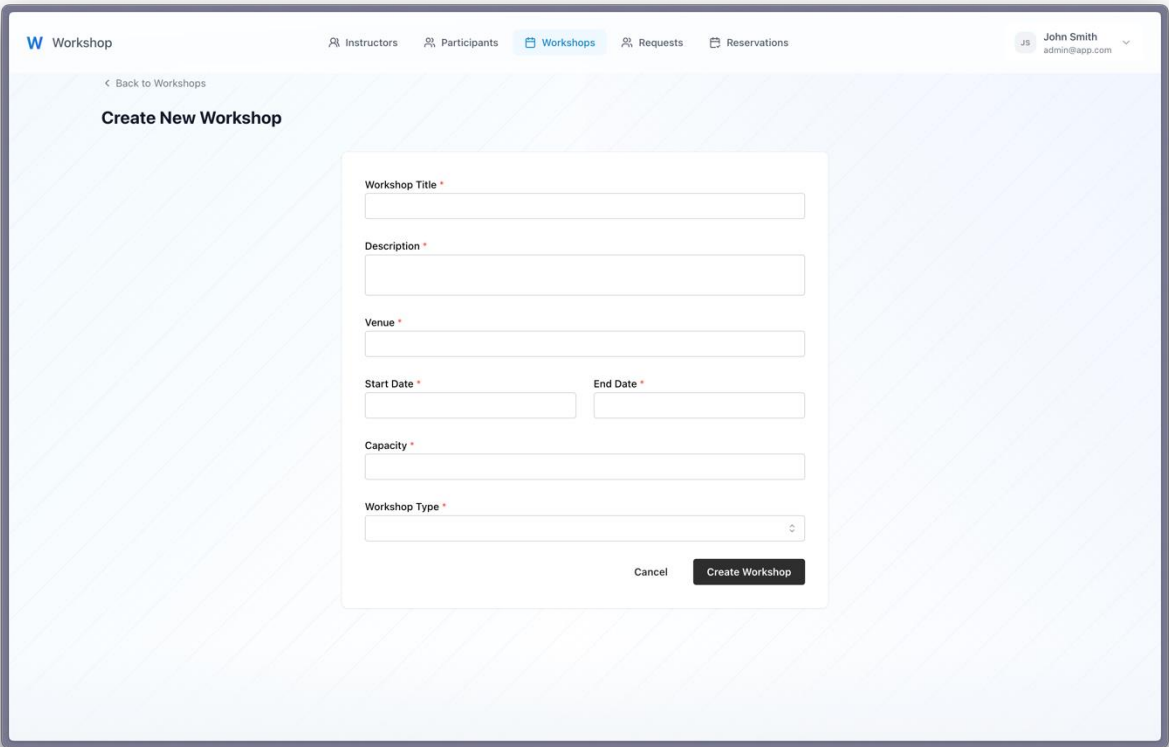
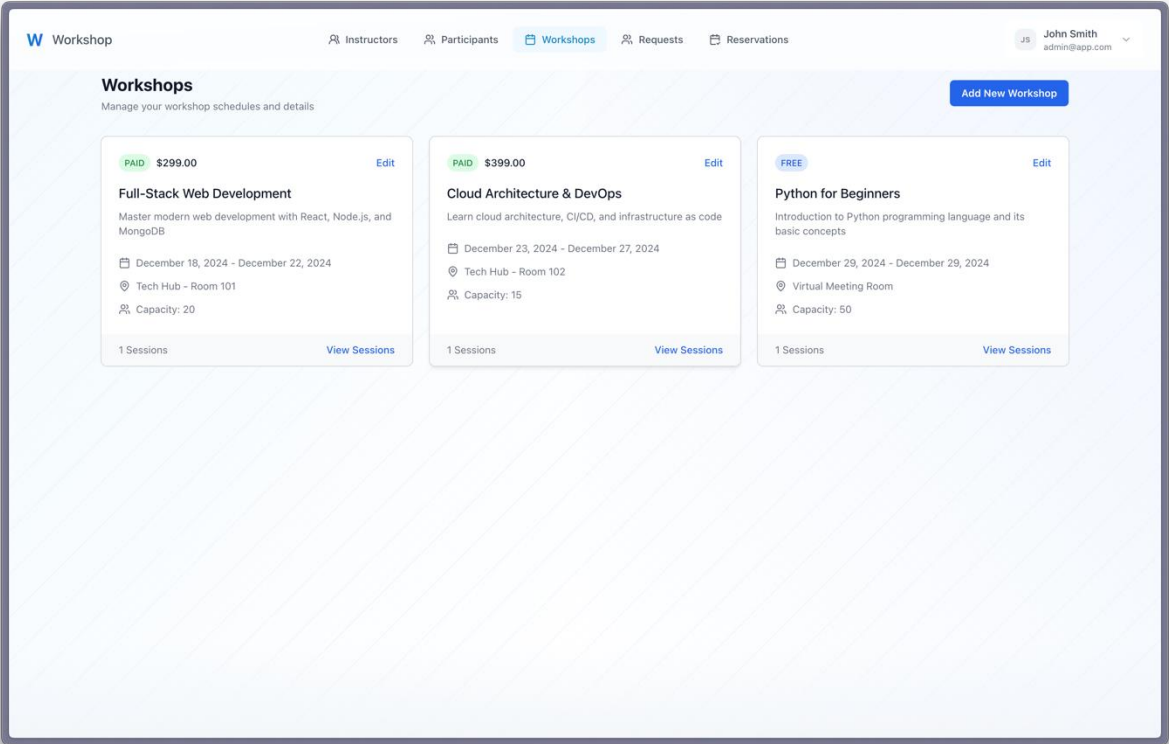
Reservations

JS John Smith
admin@app.com

Participants

View your workshop participants and their information

Name	Email	Phone	DOB	Gender
Emily Wilson	participant2@app.com	9876543210	November 19, 1988	
Vamshi Krishna	vamsikrishnayarra123@gmail.com	1234567890	March 14, 1990	



W Workshop

Instructors

Participants

Workshops

Requests

Reservations

JS John Smith
admin@app.com

< Back to Workshops

Edit Workshop

Workshop Title *

Full-Stack Web Development

Description *

Master modern web development with React, Node.js, and MongoDB

Venue *

Tech Hub - Room 101

Start Date *

Dec 18, 2024

End Date *

Dec 22, 2024

Capacity *

20

Workshop Type *

Paid

Price (USD)

299

Cancel

Update Workshop

W Workshop

Instructors

Participants

Workshops

Requests

Reservations

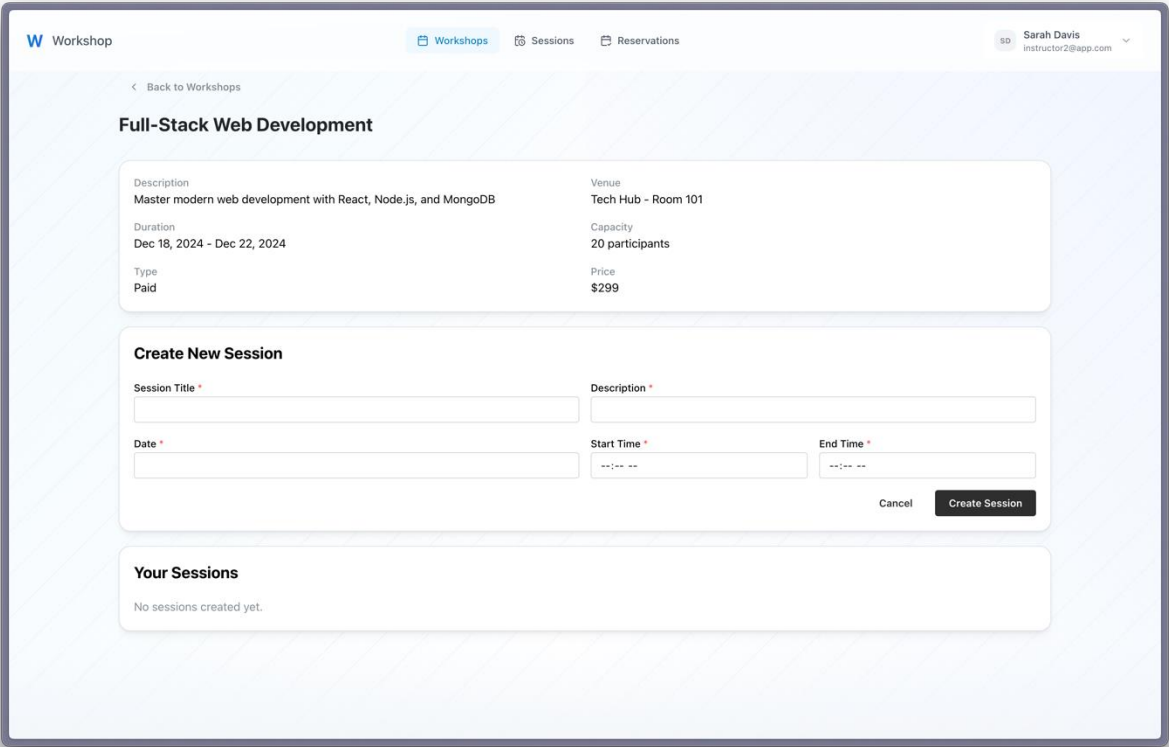
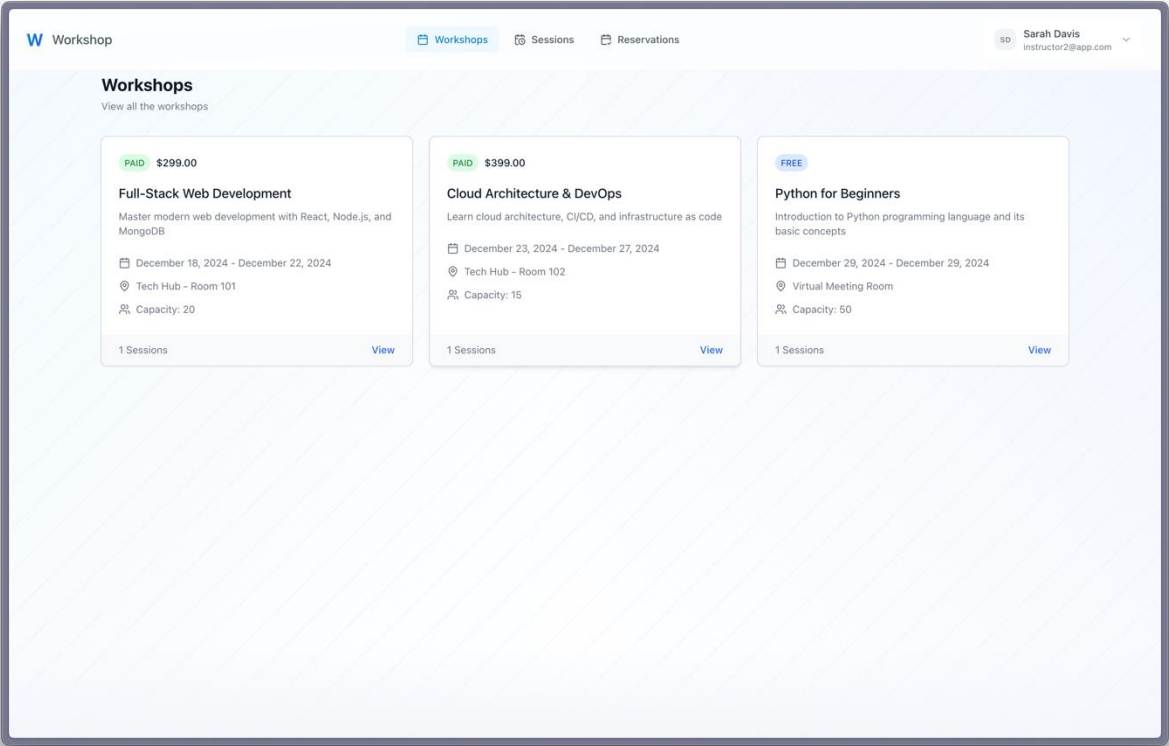
JS John Smith
admin@app.com

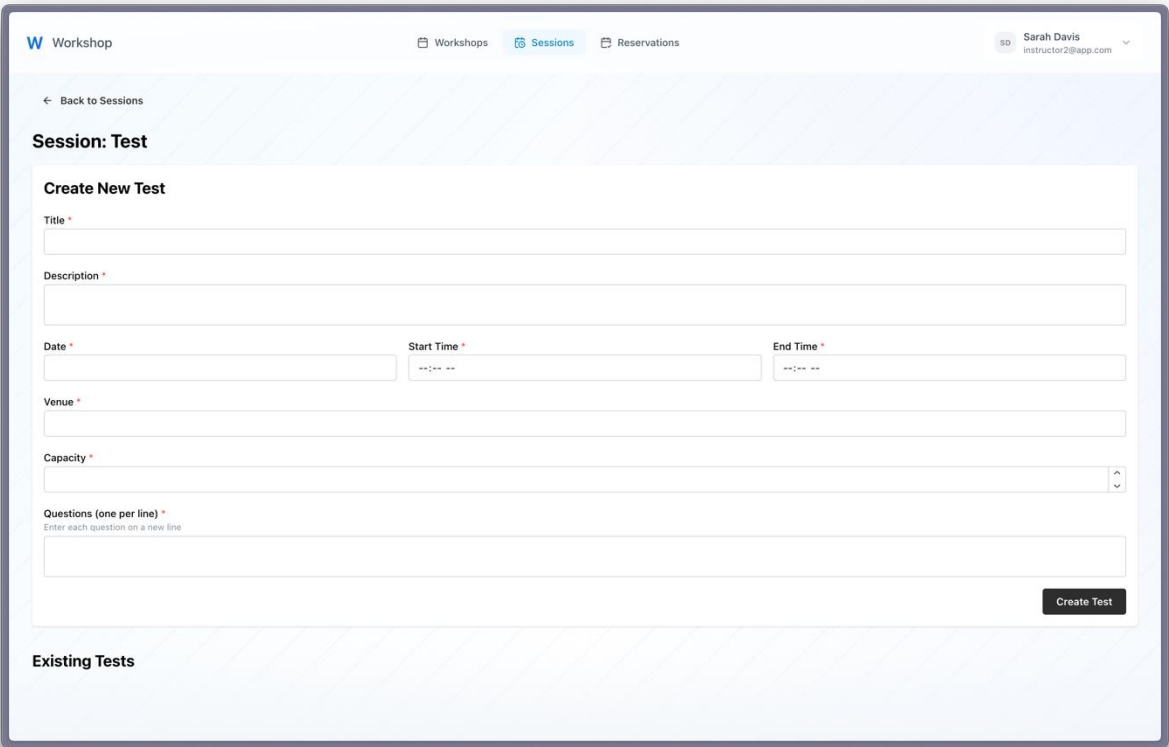
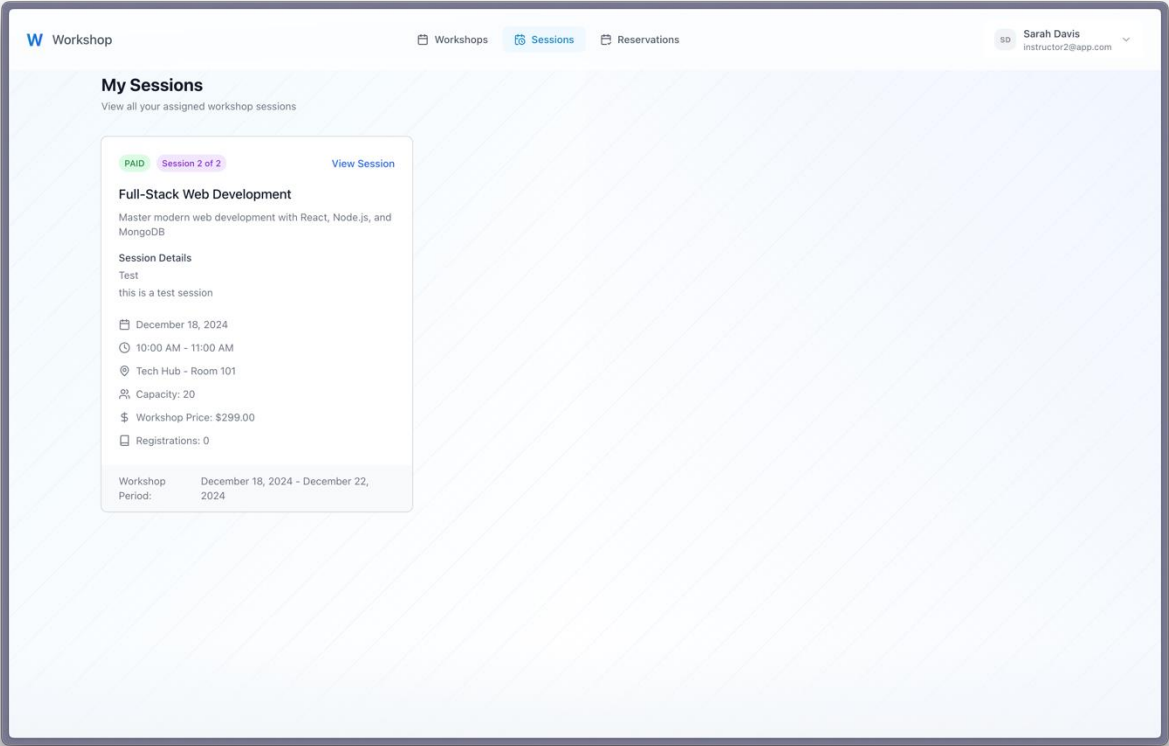
Instructor Requests

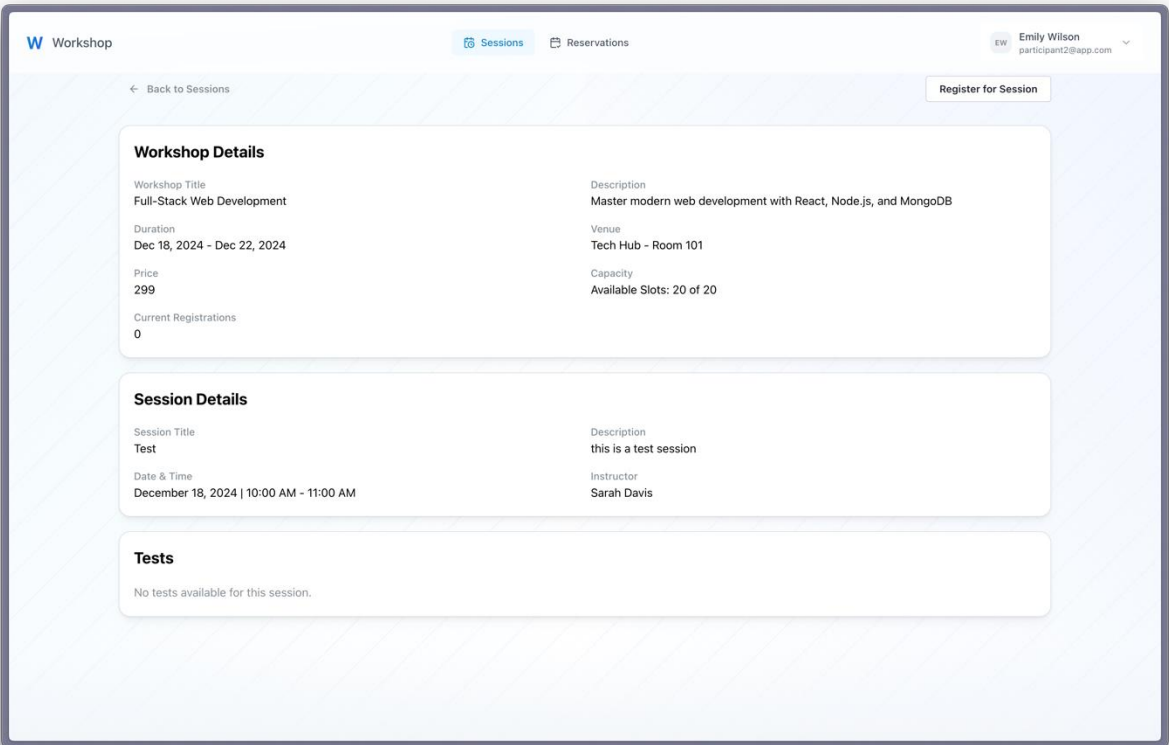
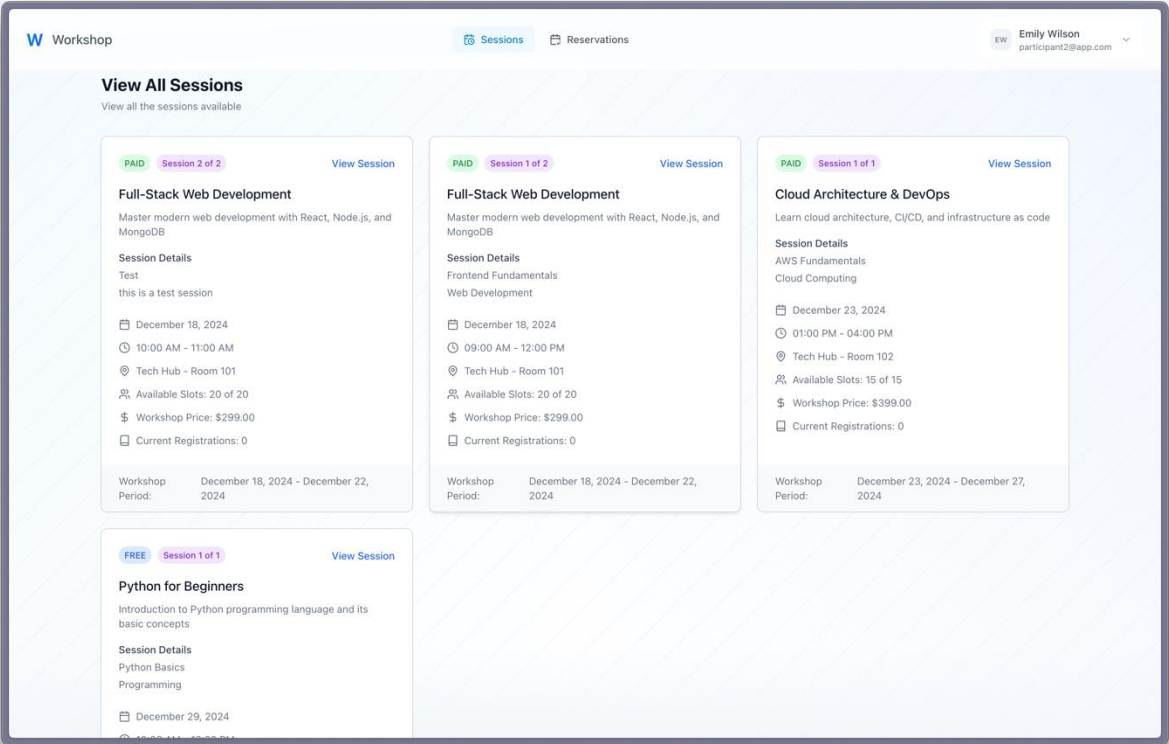
Review and manage instructor applications

No pending requests

There are currently no instructor registration requests to review.







W Workshop

SessionsReservations

VKVamshi Krishna
vamsikrishnayarra123@gmail.com

[← Back to Sessions](#)[Register for Session](#)

Workshop Details

Workshop Title

Full-Stack Web Development

Duration

Dec 18, 2024 - Dec 22, 2024

Price

299

Current Registrations

0

Description

Master modern web development with React, Node.js, and MongoDB

Venue

Tech Hub - Room 101

Capacity

Available Slots: 20 of 20

Session Details

Session Title

Test

Date & Time

December 18, 2024 | 10:00 AM - 11:00 AM

Description

this is a test session

Instructor

Sarah Davis

Tests

Test Exam

this is for testing the participants

Date

December 13, 2024

Time

10:00 AM - 11:00 AM

Venue

Test Hall

Capacity: 100 participants

W Workshop

SessionsReservations

VKVamshi Krishna
vamsikrishnayarra123@gmail.com

[← Back to Sessions](#)[Already Registered](#)

Workshop Details

Workshop Title

Full-Stack Web Development

Duration

Dec 18, 2024 - Dec 22, 2024

Price

299

Current Registrations

1

Description

Master modern web development with React, Node.js, and MongoDB

Venue

Tech Hub - Room 101

Capacity

Available Slots: 19 of 20

Session Details

Session Title

Test

Date & Time

December 18, 2024 | 10:00 AM - 11:00 AM

Description

this is a test session

Instructor

Sarah Davis

Tests

Test Exam

this is for testing the participants

Date

December 13, 2024

Time

10:00 AM - 11:00 AM

Venue

Test Hall

Capacity: 100 participants

W Workshop

SessionsReservations

VK Vamshi Krishna
vamshikrishnayarra123@gmail.com

[← Back to Reservations](#)

Registration Details

Payment Method

CASH

Amount Paid

\$299

Workshop Details

Workshop Title

Full-Stack Web Development

Description

Master modern web development with React, Node.js, and MongoDB

Duration

Dec 18, 2024 - Dec 22, 2024

Venue

Tech Hub - Room 101

Session Details

Session Title

Test

Description

this is a test session

Date & Time

December 18, 2024 || 10:00 AM - 11:00 AM

Instructor

Sarah Davis

Tests

Test Exam

Capacity: 100 participants

Register for Test

this is for testing the participants

Date

December 13, 2024

Time

10:00 AM - 11:00 AM

Venue

Test Hall

W Workshop

Test registered successfully

SessionsReservations

VK Vamshi Krishna
vamshikrishnayarra123@gmail.com

[← Back to Reservations](#)

Registration Details

Payment Method

CASH

Amount Paid

\$299

Workshop Details

Workshop Title

Full-Stack Web Development

Description

Master modern web development with React, Node.js, and MongoDB

Duration

Dec 18, 2024 - Dec 22, 2024

Venue

Tech Hub - Room 101

Session Details

Session Title

Test

Description

this is a test session

Date & Time

December 18, 2024 || 10:00 AM - 11:00 AM

Instructor

Sarah Davis

Tests

Test Exam

Capacity: 100 participants

View Questions

Complete Test

this is for testing the participants

Date

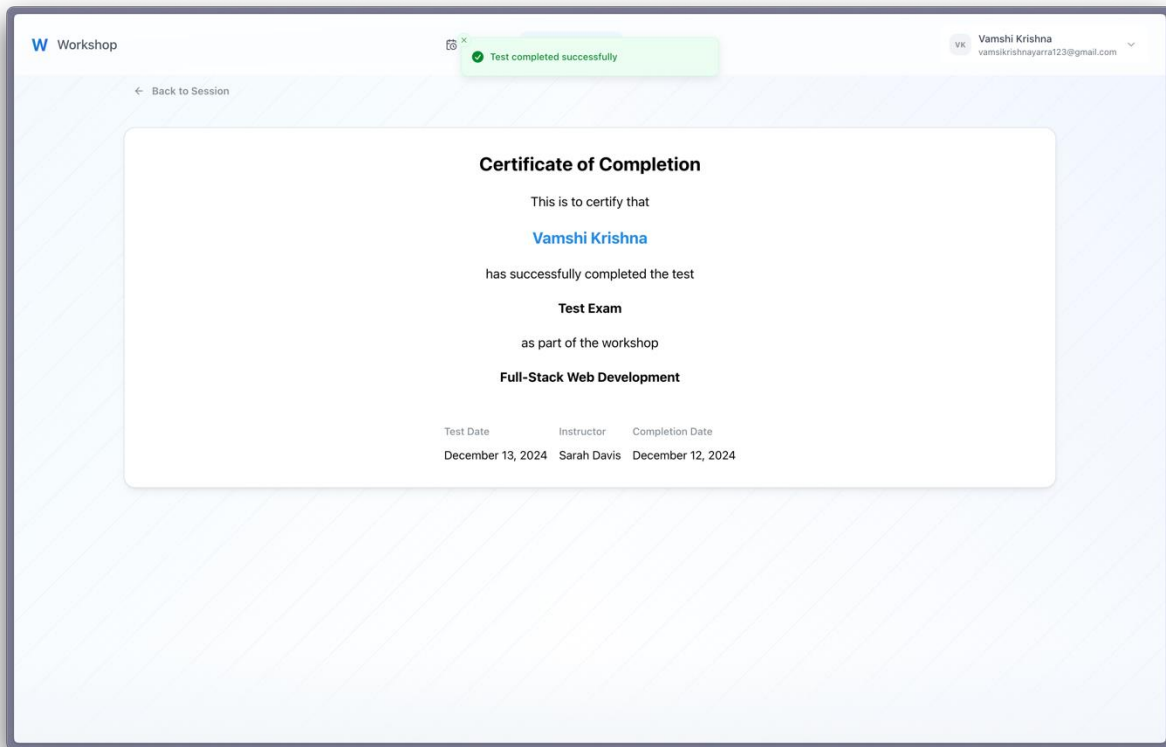
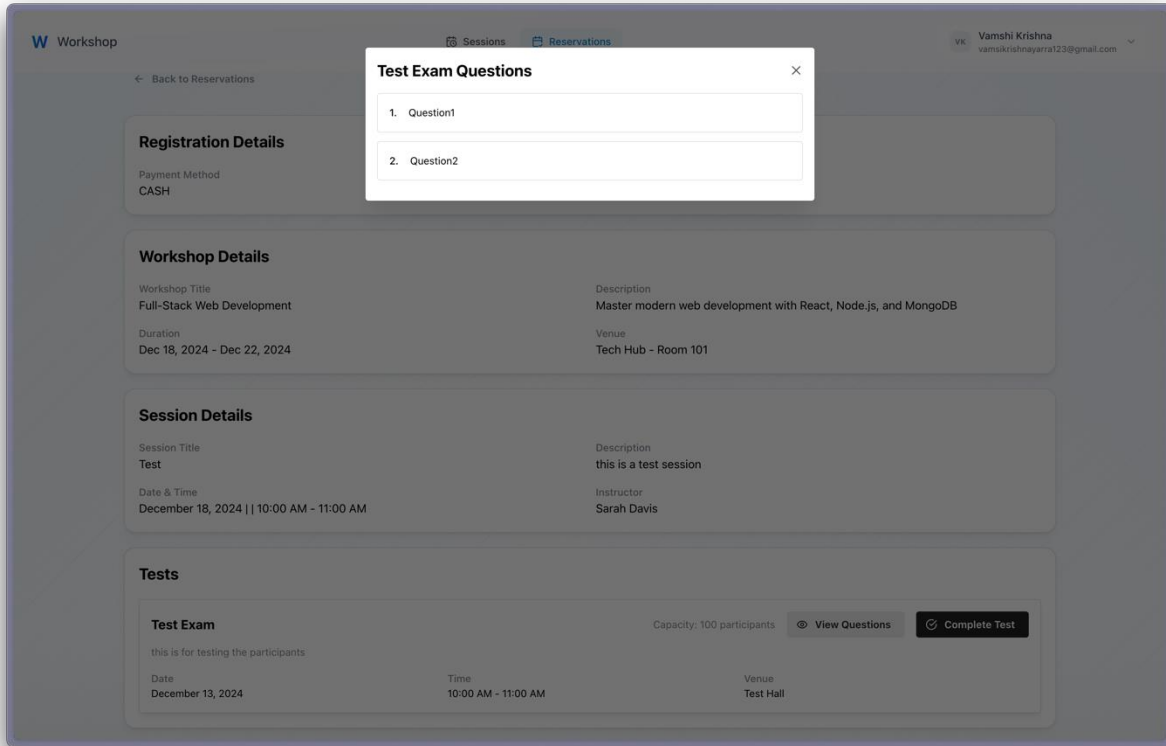
December 13, 2024

Time

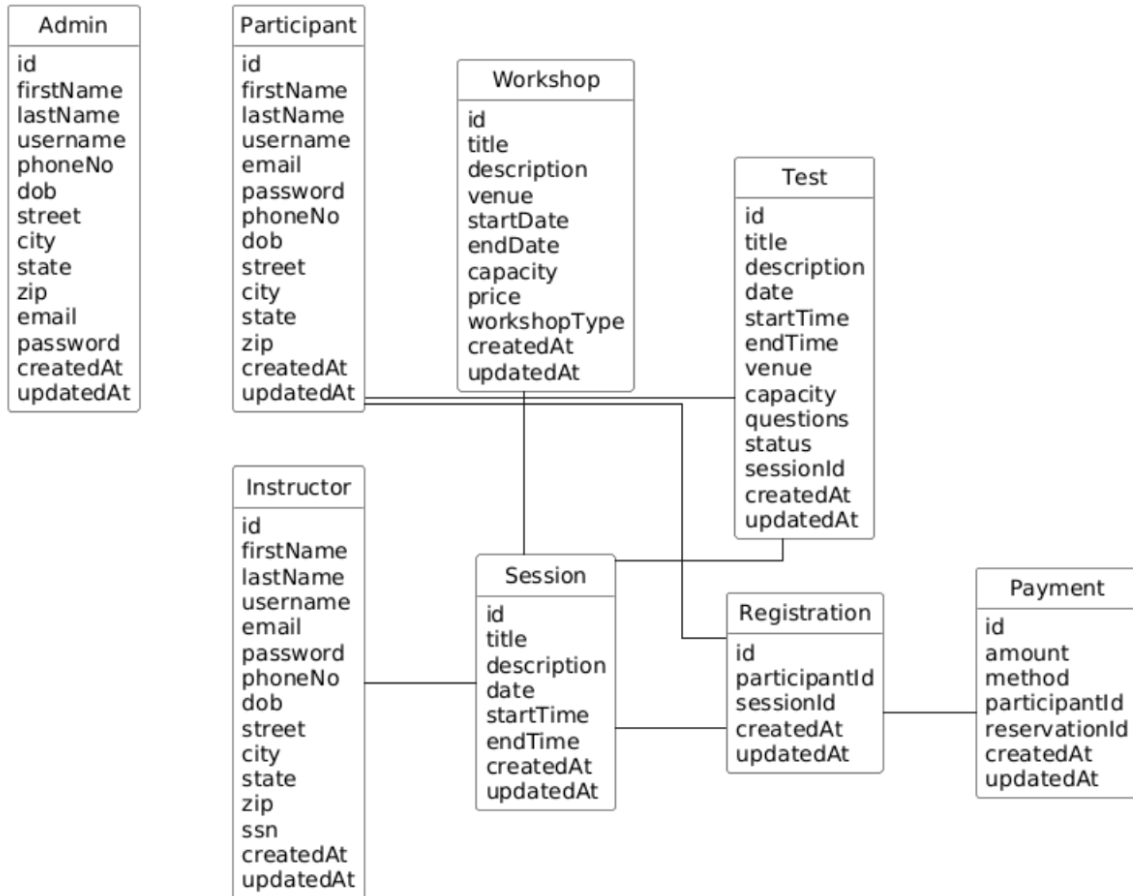
10:00 AM - 11:00 AM

Venue

Test Hall



ER Diagram



Sample Data

```
{
  "admin": [
    {
      "id": "65789a2b3c4d5e6f7g8h9i0j",
      "firstName": "John",
      "lastName": "Doe",
      "username": "johndoe_admin",
      "phoneNo": "555-0123",
      "dob": "1985-06-15T00:00:00Z",
      "street": "123 Admin Street",
      "city": "Austin",
      "state": "TX",
      "zip": "78701",
      "email": "john.doe@admin.com",
      "password": "hashedPassword123",
      "role": "ADMIN",
      "createdAt": "2024-01-01T00:00:00Z",
      "updatedAt": "2024-01-01T00:00:00Z"
    }
  ],
  "participants": [
    {
      "id": "65789a2b3c4d5e6f7g8h9i1k",
      "firstName": "Alice",
      "lastName": "Smith",
      "username": "alice_smith",
      "email": "alice.smith@email.com",
      "password": "hashedPassword456",
      "phoneNo": "555-0124",
      "dob": "1990-03-20T00:00:00Z",
      "street": "456 Participant Ave",
      "city": "Austin",
      "state": "TX",
      "zip": "78702",
      "role": "PARTICIPANT",
      "createdAt": "2024-01-02T00:00:00Z",
      "updatedAt": "2024-01-02T00:00:00Z"
    }
  ],
  "instructors": [
    {
      "id": "65789a2b3c4d5e6f7g8h9i2l",
      "firstName": "Robert",
      "lastName": "Johnson",
      "username": "rob_johnson",
      "email": "robert.johnson@instructor.com",
      "password": "hashedPassword789",
      "phoneNo": "555-0125",
      "dob": "1982-08-10T00:00:00Z",
      "street": "789 Instructor Rd",
      "city": "Austin",
      "state": "TX",
      "zip": "78703",
      "ssn": "123-45-6789",
    }
  ]
}
```

```
    "hasResetPassword": false,
    "role": "INSTRUCTOR",
    "createdAt": "2024-01-03T00:00:00Z",
    "updatedAt": "2024-01-03T00:00:00Z"
  }
],
"workshops": [
  {
    "id": "65789a2b3c4d5e6f7g8h9i3m",
    "title": "Web Development Fundamentals",
    "description": "Learn the basics of web development",
    "venue": "Tech Center Room 101",
    "startDate": "2024-02-01T00:00:00Z",
    "endDate": "2024-02-28T00:00:00Z",
    "capacity": 20,
    "price": 299.99,
    "workshopType": "PAID",
    "createdAt": "2024-01-04T00:00:00Z",
    "updatedAt": "2024-01-04T00:00:00Z"
  }
],
"sessions": [
  {
    "id": "65789a2b3c4d5e6f7g8h9i4n",
    "title": "HTML & CSS Basics",
    "description": "Introduction to HTML and CSS",
    "date": "2024-02-01T00:00:00Z",
    "startTime": "2024-02-01T09:00:00Z",
    "endTime": "2024-02-01T12:00:00Z",
    "workshopId": "65789a2b3c4d5e6f7g8h9i3m",
    "instructorId": "65789a2b3c4d5e6f7g8h9i2l",
    "createdAt": "2024-01-05T00:00:00Z",
    "updatedAt": "2024-01-05T00:00:00Z"
  }
],
"registrations": [
  {
    "id": "65789a2b3c4d5e6f7g8h9i5o",
    "participantId": "65789a2b3c4d5e6f7g8h9i1k",
    "sessionId": "65789a2b3c4d5e6f7g8h9i4n",
    "createdAt": "2024-01-06T00:00:00Z",
    "updatedAt": "2024-01-06T00:00:00Z"
  }
],
"payments": [
  {
    "id": "65789a2b3c4d5e6f7g8h9i6p",
    "amount": 299.99,
    "method": "CREDIT_CARD",
    "participantId": "65789a2b3c4d5e6f7g8h9i1k",
    "reservationId": "65789a2b3c4d5e6f7g8h9i5o",
    "createdAt": "2024-01-07T00:00:00Z",
    "updatedAt": "2024-01-07T00:00:00Z"
  }
],
"tests": [
  {
```

```

      "id": "65789a2b3c4d5e6f7g8h9i7q",
      "title": "HTML & CSS Assessment",
      "description": "Final assessment for HTML and CSS module",
      "date": "2024-02-28T00:00:00Z",
      "startTime": "2024-02-28T09:00:00Z",
      "endTime": "2024-02-28T11:00:00Z",
      "venue": "Tech Center Room 101",
      "capacity": 20,
      "questions": [
        "What is HTML?",
        "Explain CSS selectors",
        "What is the box model?"
      ],
      "status": "PENDING",
      "sessionId": "65789a2b3c4d5e6f7g8h9i4n",
      "createdAt": "2024-01-08T00:00:00Z",
      "updatedAt": "2024-01-08T00:00:00Z"
    }
  ],
  "testRegistrations": [
    {
      "id": "65789a2b3c4d5e6f7g8h9i8r",
      "participantId": "65789a2b3c4d5e6f7g8h9i1k",
      "testId": "65789a2b3c4d5e6f7g8h9i7q",
      "createdAt": "2024-01-09T00:00:00Z",
      "updatedAt": "2024-01-09T00:00:00Z"
    }
  ],
  "passwordResets": [
    {
      "id": "65789a2b3c4d5e6f7g8h9i9s",
      "participantId": "65789a2b3c4d5e6f7g8h9i1k",
      "adminId": null,
      "instructorId": null,
      "token": "reset_token_123456",
      "used": false,
      "expiresAt": "2024-01-10T00:00:00Z",
      "createdAt": "2024-01-09T00:00:00Z"
    }
  ],
  "usernameResets": [
    {
      "id": "65789a2b3c4d5e6f7g8h9i0t",
      "participantId": "65789a2b3c4d5e6f7g8h9i1k",
      "adminId": null,
      "instructorId": null,
      "token": "username_reset_123456",
      "used": false,
      "expiresAt": "2024-01-10T00:00:00Z",
      "createdAt": "2024-01-09T00:00:00Z"
    }
  ]
}

```

MongoDB Shell Commands

```
// INSERT COMMANDS
```

```
// -----

// Insert Admin
db.Admin.insertOne({
  _id: ObjectId("65789a2b3c4d5e6f7g8h9i0j"),
  firstName: "John",
  lastName: "Doe",
  username: "johndoe_admin",
  phoneNo: "555-0123",
  dob: new Date("1985-06-15"),
  street: "123 Admin Street",
  city: "Austin",
  state: "TX",
  zip: "78701",
  email: "john.doe@admin.com",
  password: "hashedPassword123",
  role: "ADMIN",
  createdAt: new Date("2024-01-01"),
  updatedAt: new Date("2024-01-01")
});

// Insert Participant
db.Participant.insertOne({
  _id: ObjectId("65789a2b3c4d5e6f7g8h9i1k"),
  firstName: "Alice",
  lastName: "Smith",
  username: "alice_smith",
  email: "alice.smith@email.com",
  password: "hashedPassword456",
  phoneNo: "555-0124",
  dob: new Date("1990-03-20"),
  street: "456 Participant Ave",
  city: "Austin",
  state: "TX",
  zip: "78702",
  role: "PARTICIPANT",
  createdAt: new Date("2024-01-02"),
  updatedAt: new Date("2024-01-02")
});

// Insert Instructor
db.Instructor.insertOne({
  _id: ObjectId("65789a2b3c4d5e6f7g8h9i2l"),
  firstName: "Robert",
  lastName: "Johnson",
  username: "rob_johnson",
  email: "robert.johnson@instructor.com",
  password: "hashedPassword789",
  phoneNo: "555-0125",
  dob: new Date("1982-08-10"),
  street: "789 Instructor Rd",
  city: "Austin",
  state: "TX",
  zip: "78703",
  ssn: "123-45-6789",
  hasResetPassword: false,
  role: "INSTRUCTOR",
});
```

```

        createdAt: new Date("2024-01-03"),
        updatedAt: new Date("2024-01-03")
    });

// Insert Workshop
db.Workshop.insertOne({
    _id: ObjectId("65789a2b3c4d5e6f7g8h9i3m"),
    title: "Web Development Fundamentals",
    description: "Learn the basics of web development",
    venue: "Tech Center Room 101",
    startDate: new Date("2024-02-01"),
    endDate: new Date("2024-02-28"),
    capacity: 20,
    price: 299.99,
    workshopType: "PAID",
    createdAt: new Date("2024-01-04"),
    updatedAt: new Date("2024-01-04")
});

// Insert Session
db.Session.insertOne({
    _id: ObjectId("65789a2b3c4d5e6f7g8h9i4n"),
    title: "HTML & CSS Basics",
    description: "Introduction to HTML and CSS",
    date: new Date("2024-02-01"),
    startTime: new Date("2024-02-01T09:00:00Z"),
    endTime: new Date("2024-02-01T12:00:00Z"),
    workshopId: ObjectId("65789a2b3c4d5e6f7g8h9i3m"),
    instructorId: ObjectId("65789a2b3c4d5e6f7g8h9i2l"),
    createdAt: new Date("2024-01-05"),
    updatedAt: new Date("2024-01-05")
});

// Insert Registration
db.Registration.insertOne({
    _id: ObjectId("65789a2b3c4d5e6f7g8h9i5o"),
    participantId: ObjectId("65789a2b3c4d5e6f7g8h9i1k"),
    sessionId: ObjectId("65789a2b3c4d5e6f7g8h9i4n"),
    createdAt: new Date("2024-01-06"),
    updatedAt: new Date("2024-01-06")
});

// Insert Payment
db.Payment.insertOne({
    _id: ObjectId("65789a2b3c4d5e6f7g8h9i6p"),
    amount: 299.99,
    method: "CREDIT_CARD",
    participantId: ObjectId("65789a2b3c4d5e6f7g8h9i1k"),
    reservationId: ObjectId("65789a2b3c4d5e6f7g8h9i5o"),
    createdAt: new Date("2024-01-07"),
    updatedAt: new Date("2024-01-07")
});

// Insert Test
db.Test.insertOne({
    _id: ObjectId("65789a2b3c4d5e6f7g8h9i7q"),
    title: "HTML & CSS Assessment",

```



```

description: "Final assessment for HTML and CSS module",
date: new Date("2024-02-28"),
startTime: new Date("2024-02-28T09:00:00Z"),
endTime: new Date("2024-02-28T11:00:00Z"),
venue: "Tech Center Room 101",
capacity: 20,
questions: [
  "What is HTML?",
  "Explain CSS selectors",
  "What is the box model?"
],
status: "PENDING",
sessionId: ObjectId("65789a2b3c4d5e6f7g8h9i4n"),
createdAt: new Date("2024-01-08"),
updatedAt: new Date("2024-01-08")
});

```

```

// READ COMMANDS
// -----

```

```

// Find all admins
db.Admin.find({});

```

```

// Find admin by email
db.Admin.findOne({ email: "john.doe@admin.com" });

```

```

// Find participant by username
db.Participant.findOne({ username: "alice_smith" });

```

```

// Find instructor with their sessions
db.Instructor.aggregate([
  {
    $match: { username: "rob_johnson" }
  },
  {
    $lookup: {
      from: "Session",
      localField: "_id",
      foreignField: "instructorId",
      as: "sessions"
    }
  }
]);

```

```

// Find workshop with all sessions
db.Workshop.aggregate([
  {
    $match: { title: "Web Development Fundamentals" }
  },
  {
    $lookup: {
      from: "Session",
      localField: "_id",
      foreignField: "workshopId",
      as: "sessions"
    }
  }
]);

```

```

]);

// Find all registrations for a participant with session details
db.Registration.aggregate([
  {
    $match: {
      participantId: ObjectId("65789a2b3c4d5e6f7g8h9i1k")
    }
  },
  {
    $lookup: {
      from: "Session",
      localField: "sessionId",
      foreignField: "_id",
      as: "session"
    }
  }
]);

// Find payments with participant and registration details
db.Payment.aggregate([
  {
    $lookup: {
      from: "Participant",
      localField: "participantId",
      foreignField: "_id",
      as: "participant"
    }
  },
  {
    $lookup: {
      from: "Registration",
      localField: "reservationId",
      foreignField: "_id",
      as: "registration"
    }
  }
]);

// Find all tests for a specific session
db.Test.find({
  sessionId: ObjectId("65789a2b3c4d5e6f7g8h9i4n")
});

// Find upcoming sessions (after current date)
db.Session.find({
  date: { $gt: new Date() }
}).sort({ date: 1 });

// Find workshops with available capacity
db.Workshop.find({
  capacity: { $gt: 0 }
});

// Find all pending tests
db.Test.find({
  status: "PENDING"
});

```

```
});  
  
// Count registrations per session  
db.Registration.aggregate([  
  {  
    $group: {  
      _id: "$sessionId",  
      totalRegistrations: { $sum: 1 }  
    }  
  }  
]);
```